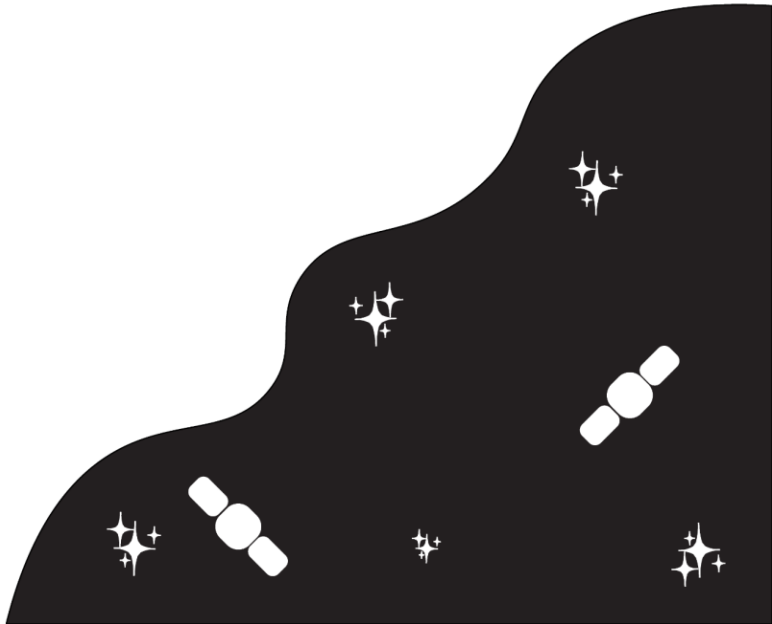
A Review of Early Excess Flux SNe Ia and Their Hosts

Riley Webster and Anais Möller

Honours Project A

Semester 1, 2026



Rubin, LSST and Fink

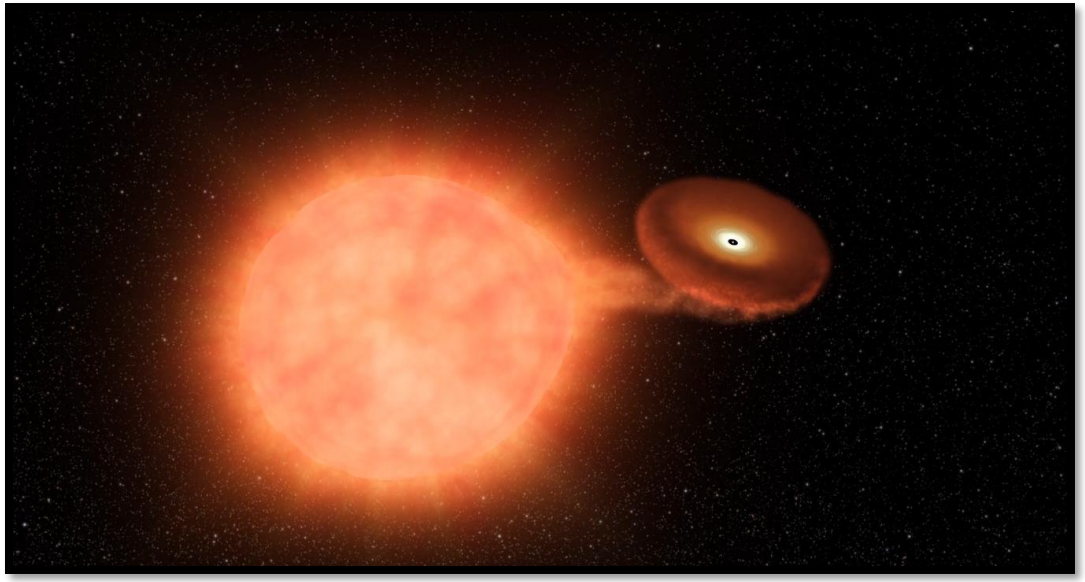
- Rubin telescope
  - 8.36 metre aperture (Bianco et al., 2022)
  - 320.5 – 1084.5 nm (*Vera C. Rubin Observatory Survey Strategy*, n.d.)
- LSST
  - Observe the southern skies.
  - Produces alerts when a change in the sky is detected.
  - Produce 10 million alerts per night, for 10 years. (Möller et al., 2021)
- Fink
  - Processes and enriches the alerts.



(Vera C. Rubin Press Release)

What are SN Ia?

- The explosion of a carbon-oxygen white dwarf (Lim et al., 2023)

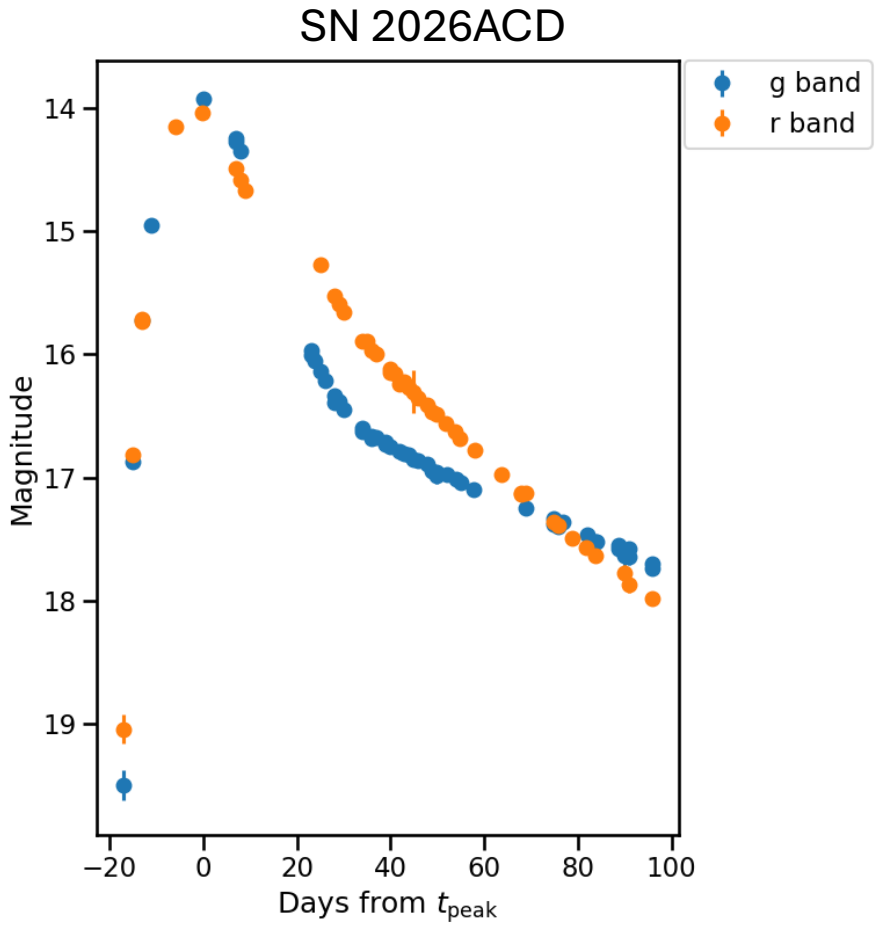


(*NASA Image and Video Library*)



SNe Ia – Why?

- SNe Ia are standardizable candles.
- $\mu = m_B^* - M_B + \alpha x_1 - \beta c$ (Tripp, 1998)
- SALT is used to fit x_1 and c (Guy et al., 2005).

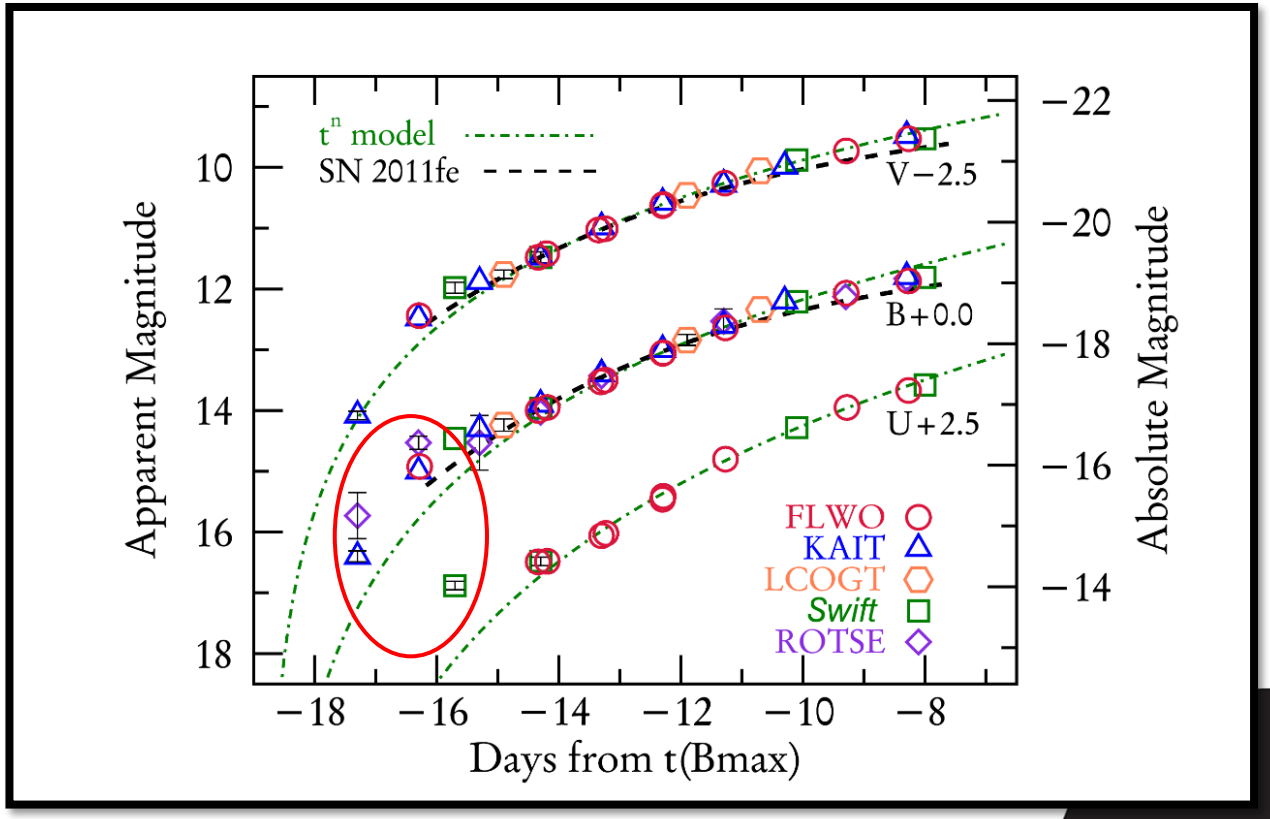


Light curve obtained from the Fink broker.

Early Excess

- In some cases, observation of SNe Ia have a bump in the light curve.

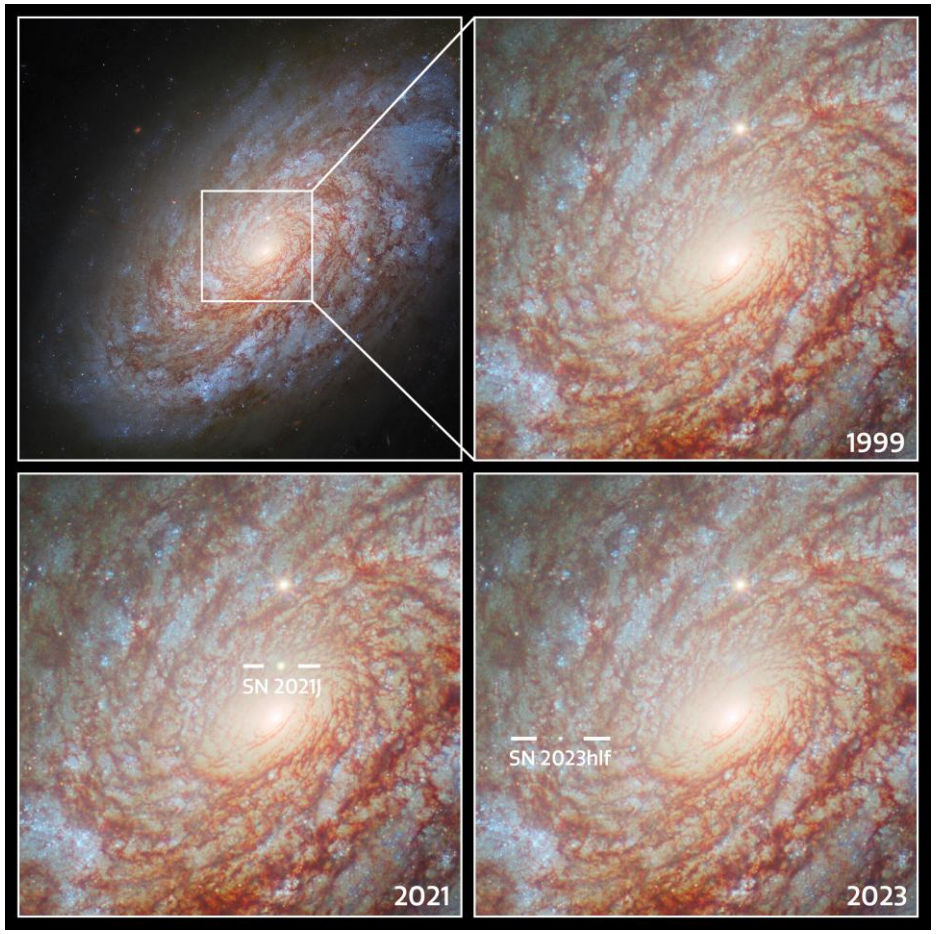
- Reasons For Excess
  - Companion interaction (Burke et al., 2022) and (Lim et al., 2023)
  - Clumping of nickel 56 and interactions with CSM (Deckers et al., 2022)



(Marion et al., 2016)

Host Galaxy Environments

- Properties
  - Mass
  - Distance
  - Star Formation Rate



(*Supernovae in NGC 4414*, n.d.)

Proposed Method

Assemble a catalogue of extragalactic candidates from Rubin using FINK.

Use ML classifiers to find potential SNe Ia.

Use SALT to fit observed light curves (LC) to determine x_1 and c.

Characterise SNe Ia using x_1 and c.

Search for excess flux by finding residuals between fits and observed LC.

Correlate SN properties with the host galaxy properties.

